The Alchemist's Archive

WORKED SOLUTION KEY

01 Chemical Reactions and Equations

MODEL ANSWERS • Match the number to the student assignment.

01 CBSE 2023 · 31/1/1 · Q2

The brown gas is nitrogen dioxide. $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$ on heating. The solid residue is lead(II) oxide.

02 CBSE 2023 · 31/1/1 · Q3

Two O atoms require $2H_2O$. That requires 4H from 4HCl. The four Cl atoms split between $MnCl_2$ and Cl_2. So $MnO_2 + 4HCl \rightarrow MnCl_2 + 2H_2O + Cl_2$.

03 CBSE 2023 · 31/1/1 · Q17

$Pb + CuSO_4 \rightarrow PbSO_4 + Cu$. Lead is more reactive than copper and displaces Cu^{2+}, reducing the blue colour. Insoluble lead sulfate can coat the strip.

04 CBSE 2023 · 31/1/1 · Q27

CO is the reducing agent. It gains oxygen to form CO_2 while Fe_2O_3 loses oxygen.

05 CBSE 2023 · 31/1/1 · Q27

NH_3 is oxidised as nitrogen goes from −3 to +2 in NO. O_2 acts as the oxidising agent and is reduced.

06 CBSE 2024 · 31/1/1 · Q2

Silver metal causes the darkening. $2AgCl \rightarrow 2Ag + Cl_2$ in sunlight. This is photochemical decomposition.

07 CBSE 2024 · 31/1/1 · Q21

$CaO + H_2O \rightarrow Ca(OH)_2$. Two reactants form one product, so it is combination. The reaction releases heat and the temperature rises.

08 CBSE 2024 · 31/1/1 · Q27

$BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$. A white insoluble solid, $BaSO_4$, appears in the mixture.

09 CBSE 2024 · 31/1/1 · Q27

$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$. Copper forms the coating. As $FeSO_4$ forms and Cu^{2+} is removed, the solution changes from blue towards pale green.

10 CBSE 2024 · 31/1/1 · Q7

Mn drops from +4 in MnO_2 to +2 in $MnCl_2$ and is reduced. Some Cl rises from −1 in chloride to 0 in Cl_2 and is oxidised.

11 CBSE 2025 · 31/1/1 · Q1

$2H_2O \rightarrow 2H_2 + O_2$. From 36 g of water, 4 g is hydrogen and 32 g is oxygen. Check that 4 + 32 = 36 g.

12 CBSE 2025 · 31/1/1 · Q1

The equation $2H_2O \rightarrow 2H_2 + O_2$ gives a 2:1 volume ratio. Hydrogen is 2 × 18 = 36 mL. It is collected at the cathode.

13 CBSE 2025 · 31/1/1 · Q2

$2Mg + O_2 \rightarrow 2MgO$, magnesium oxide. $4Al + 3O_2 \rightarrow 2Al_2O_3$, aluminium oxide.

14 CBSE 2025 · 31/1/1 · Q17

Composting releases heat as microbes break down organic matter. It is a counterexample. Many other decomposition reactions require an input of energy, such as heating lead nitrate.

15 CBSE 2025 · 31/1/1 · Q21

Dissolving can be a physical change. Gas evolution or a newly formed precipitate can suggest reaction. Stronger confirmation comes from identifying new substances.

16 CBSE 2026 · 31/1/1 · Q17

$AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$. Remove spectator Na^+ and NO_3^- to obtain $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$.

17 CBSE 2026 · 31/1/1 · Q19

$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$. Two formula units give four NO_2 molecules and one O_2 molecule.

18 CBSE 2026 · 31/1/1 · Q26

ZnO is the oxidising agent. It supplies oxygen to carbon, forming CO, while ZnO itself is reduced to zinc.

19 CBSE 2026 · 31/1/1 · Q26

$Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$. The insoluble yellow solid is lead(II) iodide.

20 CBSE 2026 · 31/1/1 · Q26

Water requires electrical energy, $2H_2O \rightarrow 2H_2 + O_2$. Silver chloride decomposes in light, $2AgCl \rightarrow 2Ag + Cl_2$.

21 CBSE 2024 · 31/1/1 · Q27

$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$. Na^+ and Cl^- remain in solution and are spectator ions.

22 CBSE 2025 · 31/1/1 · Q2

$4Al + 3O_2 \rightarrow 2Al_2O_3$. The required oxygen is $0.80 \times 3/4 = 0.60$ mol.

23 CBSE 2024 · 31/1/1 · Q21

$CaO + H_2O \rightarrow Ca(OH)_2$. The mole ratio is 1:1. 56 g CaO is 1 mol. One mole of $Ca(OH)_2$ has mass $40 + 2(16 + 1) = 74$ g.

24 CBSE 2023 · 31/1/1 · Q27

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$. The Fe:CO mole ratio is 2:3, so iron formed is $0.60 \times 2/3 = 0.40$ mol.

25 CBSE 2024 · 31/1/1 · Q2

$2AgCl \rightarrow 2Ag + Cl_2$. Silver is formed in a 1:1 ratio with AgCl, so 0.30 mol Ag forms. Chlorine is formed in a 1:2 ratio, so 0.15 mol Cl_2 forms.

02 Acids, Bases and Salts

26 CBSE 2023 · 31/1/1 · Q1

$Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$. Then $CO_2 + Ca(OH)_2 \rightarrow CaCO_3\downarrow + H_2O$. Calcium carbonate makes the limewater milky.

27 CBSE 2023 · 31/1/1 · Q1

The insoluble calcium carbonate reacts with extra CO_2 and water to give soluble calcium hydrogen carbonate. $CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$.

28 CBSE 2023 · 31/1/1 · Q4

It is basic because pH > 7. It turns red litmus blue.

29 CBSE 2023 · 31/1/1 · Q6

Washing soda is $Na_2CO_3 \cdot 10H_2O$. The crystals contain ten waters per formula unit. Anhydrous sodium carbonate is Na_2CO_3.

30 CBSE 2023 · 31/1/1 · Q21

Blue hydrated copper sulfate is $CuSO_4 \cdot 5H_2O$. On heating, $CuSO_4 \cdot 5H_2O \rightarrow CuSO_4 + 5H_2O$. The white anhydrous salt rehydrates and turns blue.

31 CBSE 2023 · 31/1/1 · Q21

Baking soda is $NaHCO_3$. Baking powder mixes $NaHCO_3$ with a mild edible acid. CO_2 bubbles expand in the dough and make it rise.

32 CBSE 2023 · 31/1/1 · Q21

$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$. The $NaHCO_3$:CO_2 ratio is 2:1, so 0.20 mol CO_2 forms.

33 CBSE 2023 · 31/1/1 · Q28

The pH falls. Bacteria produce lactic acid from milk sugars, increasing the concentration of hydronium ions.

34 CBSE 2023 · 31/1/1 · Q28

A base neutralises some excess HCl. For magnesium hydroxide, $Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$.

35 CBSE 2024 · 31/1/1 · Q5

$Na_2CO_3 \cdot 10H_2O$ contains 10 mol water per mole of crystals. The mass is $10 \times 18 = 180$ g.

36 CBSE 2024 · 31/1/1 · Q28

The water becomes more acidic as hydronium concentration increases. Aquatic species tolerate a limited pH range, and the change can impair survival or reproduction.

37 CBSE 2024 · 31/1/1 · Q28

A strong acid ionises essentially completely, producing more hydronium ions. A weak acid ionises only partly, so more of its molecules remain un-ionised.

38 CBSE 2024 · 31/1/1 · Q28

pH 3 is more acidic. $[H_3O^+]$ ratio = $10^{-3} / 10^{-5} = 100$. The pH 3 solution has 100 times the hydronium concentration.

39 CBSE 2025 · 31/1/1 · Q3

13 < 7 < 5 < 3 in acidity. Lower pH corresponds to higher hydronium concentration.

40 CBSE 2025 · 31/1/1 · Q4

$MgO + 2HCl \rightarrow MgCl_2 + H_2O$. A basic oxide and an acid react to form salt and water.

41 CBSE 2025 · 31/1/1 · Q4

$CO_2 + 2NaOH \rightarrow Na_2CO_3 + H_2O$. CO_2 behaves as an acidic oxide because it reacts with the base.

42 CBSE 2026 · 31/1/1 · Q22

Oxalic acid occurs in tomato, lactic acid in curd, and methanoic acid in nettle stings.

43 CBSE 2026 · 31/1/1 · Q25

Na_2O is basic, CO_2 is acidic, and ZnO is amphoteric. ZnO reacts with both acids and strong bases.

44 CBSE 2026 · 31/1/1 · Q27a

$2NaCl + 2H_2O \rightarrow 2NaOH + Cl_2 + H_2$. Chlorine forms at the anode and hydrogen at the cathode.

45 CBSE 2026 · 31/1/1 · Q27a

$2NaCl + 2H_2O \rightarrow 2NaOH + Cl_2 + H_2$. NaOH is in a 1:1 ratio with NaCl, giving 0.60 mol. Cl_2 is in a 1:2 ratio, giving 0.30 mol.

46 CBSE 2026 · 31/1/1 · Q27b

Chlorine is passed over dry slaked lime. $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$.

47 CBSE 2026 · 31/1/1 · Q27b

$2CaSO_4 \cdot 2H_2O \rightarrow 2CaSO_4 \cdot \frac{1}{2}H_2O + 3H_2O$ on controlled heating near 373 K.

48 CBSE 2023 · 31/1/1 · Q21

The hemihydrate rehydrates to gypsum. $2CaSO_4 \cdot \frac{1}{2}H_2O + 3H_2O \rightarrow 2CaSO_4 \cdot 2H_2O$. Interlocking gypsum crystals harden the cast.

49 CBSE 2026 · 31/1/1 · Q27b

$NaCl + H_2O + CO_2 + NH_3 \rightarrow NaHCO_3 + NH_4Cl$. The other product is ammonium chloride.

50 CBSE 2025 · 31/1/1 · Q4

$HCl + NaOH \rightarrow NaCl + H_2O$. The coefficients give a 1:1 mole ratio, so 0.25 mol HCl is required.

03 Metals and Non-metals

51 CBSE 2023 · 31/1/1 · Q5

Aluminium is the reducing metal. It forms Al_2O_3, an amphoteric oxide that reacts with both acids and strong bases.

52 CBSE 2023 · 31/1/1 · Q37

The coating is copper(II) oxide. $2Cu + O_2 \rightarrow 2CuO$.

53 CBSE 2023 · 31/1/1 · Q37

$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$. In the school-level representation, $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$.

54 CBSE 2024 · 31/1/1 · Q1

$Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$. Hydrogen is the gas and sodium zincate is the salt.

55 CBSE 2024 · 31/1/1 · Q3

Mercury is a liquid metal. Bromine is a liquid non-metal.

56 CBSE 2024 · 31/1/1 · Q17

Dilute HCl allows hydrogen gas to escape. Nitric acid acts as an oxidising agent and generally oxidises the hydrogen formed to water while being reduced to nitrogen oxides.

57 CBSE 2024 · 31/1/1 · Q37

Impure copper is the anode, a thin sheet of pure copper is the cathode, and acidified copper sulfate solution is the electrolyte.

58 CBSE 2024 · 31/1/1 · Q37

Copper ions take electrons at the cathode, $Cu^{2+} + 2e^{-} \rightarrow Cu$, and deposit as copper. Insoluble impurities collect below the anode as anode mud.

59 CBSE 2025 · 31/1/1 · Q5

Ions are fixed in a lattice in the solid. In the molten compound they are mobile and carry charge to the electrodes.

60 CBSE 2025 · 31/1/1 · Q6

Use electrolysis of molten magnesium chloride. Magnesium is too reactive for carbon to reduce its stable oxide under usual extraction conditions.

61 CBSE 2025 · 31/1/1 · Q7

Mg loses $2e^-$ to form Mg^{2+}. O gains $2e^-$ to form O^{2-}, completing its outer shell. Charge balance gives MgO.

62 CBSE 2025 · 31/1/1 · Q27a

Calcination converts $ZnCO_3 \rightarrow ZnO + CO_2$. The oxide is easier to reduce to zinc than the original carbonate.

63 CBSE 2025 · 31/1/1 · Q27a

This is roasting. $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$. Oxygen converts the sulfide into an oxide.

64 CBSE 2025 · 31/1/1 · Q28a

The nail in aerated water rusts. Dry air lacks water. Boiling removes dissolved oxygen, and the oil limits its return. Rusting requires both water and oxygen.

65 CBSE 2026 · 31/1/1 · Q21

Graphite conducts electricity because it has delocalised electrons that can move within its layers.

66 CBSE 2026 · 31/1/1 · Q23

$Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$. Bubbles of hydrogen adhere to the calcium and make it float.

67 CBSE 2026 · 31/1/1 · Q28

Partial roasting: $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$. Self-reduction: $2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$. The remaining sulfide reduces the oxide.

68 CBSE 2026 · 31/1/1 · Q28

Solder has a low melting point. It can melt and flow between the wire ends while the higher-melting wire material stays solid.

69 CBSE 2026 · 31/1/1 · Q28

$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$. Each mole of oxide needs 2 mol Al and gives 2 mol Fe. Hence 1.00 mol Al is used and 1.00 mol Fe forms.

70 CBSE 2026 · 31/1/1 · Q28

$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$. Each mole of oxide gives 3 mol Cu, so $0.40 \times 3 = 1.20$ mol copper.

71 CBSE 2025 · 31/1/1 · Q5

Oppositely charged ions attract strongly throughout the lattice. A large energy input is needed to overcome enough attractions for the ions to move.

72 CBSE 2025 · 31/1/1 · Q6

Magnesium forms at the cathode, $Mg^{2+} + 2e^- \rightarrow Mg$. Chlorine forms at the anode when chloride ions lose electrons.

73 CBSE 2024 · 31/1/1 · Q37

The impure plate needs to be the anode so its copper can oxidise and enter solution. With the connections swapped, copper deposits on the impure plate instead.

74 CBSE 2025 · 31/1/1 · Q28a

The paint is a barrier that limits contact with both oxygen and water. If scratched through to iron, the exposed area can rust.

75 CBSE 2026 · 31/1/1 · Q28

$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$. One mole of Fe_2O_3 needs 2 mol Al, or $2 \times 27 = 54$ g.

04 Carbon and Its Compounds

76 CBSE 2022 · 31/1/1 · Q1

Methane is CH_4, with four C–H bonds. Propane is C_3H_8, represented as $CH_3–CH_2–CH_3$.

77 CBSE 2022 · 31/1/1 · Q1

Boiling a simple molecular compound overcomes relatively weak intermolecular forces. In an ionic salt strong electrostatic attractions must be overcome.

78 CBSE 2022 · 31/1/1 · Q9a

Butane: $CH_3–CH_2–CH_2–CH_3$. 2-methylpropane: $CH_3–CH(CH_3)–CH_3$. Both are C_4H_{10}, with straight and branched carbon skeletons respectively.

79 CBSE 2022 · 31/1/1 · Q9b

Benzene is C_6H_6. The alternating sketch has 3 C=C bonds, 3 single C–C bonds and 6 single C–H bonds, or 9 single bonds in total.

80 CBSE 2022 · 31/1/1 · Q9b

The formula is C_nH_{2n-2}. For n = 5, the number of hydrogens is $10 - 2 = 8$. The formula is C_5H_8.

81 CBSE 2023 · 31/1/1 · Q7



Cyclohexane C_6H_{12}



Cyclohexene C_6H_{10}

Cyclohexane has six CH_2 units. Cyclohexene has two CH units and four CH_2 units.

82 CBSE 2023 · 31/1/1 · Q34

For an alkene C_nH_{2n}, mass = 14n. Thus n = 2. The alkene is ethene, C_2H_4. The alcohol that loses H_2O to give it is ethanol, C_2H_5OH.

83 CBSE 2023 · 31/1/1 · Q34

$CH_2{=}CH_2 + H_2 \rightarrow CH_3{-}CH_3$ with nickel catalyst. Ethene is hydrogenated to ethane.

84 CBSE 2024 · 31/1/1 · Q4

It lacks mobile charged particles such as ions or free electrons, so current is not readily carried through the liquid.

85 CBSE 2024 · 31/1/1 · Q34a

They share a functional group and therefore show similar characteristic reactions. Each successive member adds CH_2, which generally increases intermolecular attraction and boiling point.

86 CBSE 2024 · 31/1/1 · Q34a

One step adds CH_2. The relative molecular mass increases by $12 + 2 \times 1 = 14$.

87 CBSE 2024 · 31/1/1 · Q34a

Propanal is $CH_3–CH_2–CHO$, with the carbonyl at the end. Propanone is $CH_3–CO–CH_3$, with the carbonyl between two carbon groups.

88 CBSE 2024 · 31/1/1 · Q34b

$CH_3CH_2OH \rightarrow CH_2=CH_2 + H_2O$, using excess concentrated H_2SO_4 at about 443 K. The acid helps remove water.

89 CBSE 2025 · 31/1/1 · Q34a

$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$. Three moles of oxygen are used for each mole of ethanol.

90 CBSE 2025 · 31/1/1 · Q34a

$CH_3CH=CH_2 + H_2 \rightarrow CH_3CH_2CH_3$ over Ni. The ratio is 1:1. The hydrogen needed is 0.50 mol × 2 g/mol = 1.0 g.

91 CBSE 2025 · 31/1/1 · Q34b

Sodium ethoxide and hydrogen form. $2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$.

92 CBSE 2025 · 31/1/1 · Q34b

$2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$. The ethanol-to-hydrogen mole ratio is 2:1. Therefore 0.15 mol H_2 is formed.

93 CBSE 2026 · 31/1/1 · Q20

The group is –CHO. Its carbonyl carbon is at the end and bears a hydrogen. CH_3CHO is ethanal.

94 CBSE 2026 · 31/1/1 · Q24

Loss of four electrons takes a great deal of energy, and gaining four is difficult for a small atom. Sharing enables carbon to fill its valence shell and form covalent bonds.

95 CBSE 2026 · 31/1/1 · Q29a

Calcium and magnesium ions form insoluble salts with soap anions. The precipitate uses up soap, reducing lather.

96 CBSE 2026 · 31/1/1 · Q29a

Carbon forms many long chains because C–C bonds are strong. Si–Si bonds are weaker, so silicon shows much less catenation.

97 CBSE 2026 · 31/1/1 · Q29a

$CH_3CH_2OH + 2[O] \rightarrow CH_3COOH + H_2O$. Warm with acidified potassium dichromate or an appropriate permanganate oxidising solution.

98 CBSE 2026 · 31/1/1 · Q29b

$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$. The reaction is esterification. Concentrated H_2SO_4 acts as an acid catalyst and helps remove water.

99 CBSE 2026 · 31/1/1 · Q29b

$CH_3COOC_2H_5 + NaOH \rightarrow CH_3COONa + C_2H_5OH$. This is alkaline hydrolysis or saponification, yielding sodium ethanoate and ethanol.

100 CBSE 2026 · 31/1/1 · Q29b

Acid and ester are in a 1:1 mole ratio. The theoretical amount is 0.20 mol ester. The maximum mass is $0.20 \times 88 = 17.6$ g.

The reference shelf

Each citation identifies the source idea in a CBSE Class 10 Science board paper. Prompts are paraphrased, split into individual skills, or extended with fresh data. Question labels refer to the source paper, and worksheet numbers refer to this assignment.

2022 · Science 086 · Paper 31/1/1 · Term II

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